

Task 2: Housekeeping Telemetry and Task Scheduling

Assume that the payload software periodically monitors subsystem health while simultaneously processing GNSS measurements. Design:

- A housekeeping telemetry packet suitable for nominal operations as per CCSDS norms.
- A software scheduling strategy for all periodic and event-driven tasks.

Clearly justify:

- Task priorities
- Scheduling methodology
- Housekeeping update rates
- Event-driven versus periodic telemetry

② To address the designing of hk-telemetry packet as per CCSDS norms, one needs clear understanding of CCSDS protocol and is not possible within given timeframe.

③ → software scheduling strategy;

There are many strategies that can be implemented for scheduling tasks. For eg: one can choose shortest task first, fixed amount of every task, and others. However, these strategies may not be ideal when a critical task needs priority. For these tasks, it is best to go with preddefined priority tasks with respective codes.

"If there is a need to run a high-priority task, then pause the low-priority task temporarily"

Considering this approach, let me define the high level tasks as mentioned in the table below.

Priority	Task	Event type	Frequency	Operation.
1 (Highest)	Reset watchdog	periodic	10 Hz	Prevents OBC from freeze.
2	GNSS_data ingest	Event based	Every time data arrives	Collect incoming GPS data.
3	Health monitor	periodic	1 Hz	Reads Hardware sensors.
4	GNSS_process	Event-based	1 Hz	OD and associated calculation
5 (Lowest)	House-keeping	periodic	Every 10 sec	Gather all data & send it to ground.
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Note: - The above list is just an example. One can elaborate it and stack as many tasks possible.

(.) The frequency mentioned are arbitrary.

(.) This approach gives the flexibility to scale the list without modifying the workflow.

→ The watchdog task is prioritized highest just to make sure the system freezes are accounted. Then GNSS-data-ingestion is ranked so as to collect high-rate data. The heavy math task GNSS process is pushed to a lowest priority so that it doesn't block critical health checks.

→ Time-based scheduling is used for task that occurs at specific intervals. Event-based scheduling for data processing tasks. This approach reduces CPU load.

→ Regarding hk update rules, currently it is proposed to update & relay telemetry every 10 seconds. This helps ground operators to analyse the telemetry & also not overloading the radio bandwidth.

→ As mentioned before, all periodic-task telemetry as sent at regular interval. However, if any event-based telemetry is triggered, then it will be prioritized & sent instantly, bypassing the standard queue.

